

Testing of Hypothesis: A test procedure used to decide whether a hypothesis is to be rejected or not is called testing of hypothesis.

Making Decision: In the process of hypothesis testing, results are interpreted at the final stage. For this purpose, we compare the computed value of a test statistic with the following way:

Compare the computed value of test statistic with the table value at a predetermined level of significance. If the computed value is greater than the table value of 5% or 1% level of significance, then null hypothesis is rejected and if computed value is less than the table value of 5% or 1% level of significance, then null hypothesis is accepted.

Hypothesis test

To make the generalization about the population from the sample, statistical tests are used. These hypothetical testing related to differences are classified as parametric and nonparametric tests.

In the literal meaning of the terms, a parametric statistical test is one that makes assumptions about the parameters (defining properties) of the population distribution(s) from which one's data are drawn, while a non-parametric test is one that makes no such assumptions.

Parametric Testing

Z-Statistic (for large samples) for testing of

- single population mean,
- difference of two population means,
- single population proportion,
- difference of two population proportions.

t-Statistic (for small samples) for testing of

- single population mean,
- difference of two population means,
- paired data,
- correlation coefficient.
- **F-Statistic:** Testing equality of two population variances.
- **Chi-square:** Testing hypothetical variance

Non-parametric Testing

- Sign test: Testing median
- Wilcoxon signed rank test: Testing median
- Run test: For randomness
- Median test: For testing equality of two medians
- Mann-Whitney test: For testing equality of two medians
- Chi-square test: Testing “goodness of fit”, testing independence, homogeneity
- Kruskal Wallis: One way ANOVA

t-test

t-test is small sample test (sample size is less than 30). It was developed by William Gosset in 1908. He published this test under the name of "Student". Therefore it is known as Student's t-test. t-test is defined as the deviation of estimated mean expressed in terms standard error.

Hypothesis Testing using t-test

Testing for single population mean

Hypothesis:

- H_0 : There is no significant difference between the sample mean and population mean.
- $H_0: \mu = \mu_0$
- H_0 : The given random sample drawn from a normal population.

Assumptions:

- The sample must be drawn from a normal population, i.e., $N(\mu, \sigma^2)$.
- ~~The sample size must be less than 30.~~ The sample is random (observations are independent).
- The population variance (σ^2) is unknown.

Test-Statistic:

Under the null hypothesis (H_0), the test-statistic is

$$t = \frac{\bar{y} - \mu_0}{S / \sqrt{n}} = \frac{\bar{y} - \mu_0}{\sqrt{s^2/(n-1)}} \quad \because \quad S^2 = \frac{s^2 \cdot n}{(n-1)}$$

Where n = sample size, $S = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}}$ S is the modified standard deviation of the sample and μ_0 is the specified value of the population mean. $[(n-1) \text{ degree of freedom}]$

State the level of significance (α).

Interpretation: Based on Numerical computation:

$$\begin{array}{ll} t_{\text{cal}} < t_{\text{tab}} & \text{Accept } H_0 \\ t_{\text{cal}} > t_{\text{tab}} & \text{Reject } H_0 \end{array}$$

Q. A machinist is making parts with axle diameter of 0.700 inches. A random sample of 10 parts shows a mean diameter of 0.742 inch with a S.D. of 0.040 inch. Compute the statistic you would use to test whether the work is meeting the specification. Also state how you would proceed further.

Solⁿ. $\mu = 0.700$ inch ; $\bar{x} = 0.742$ inch $s = 0.040$ inch

and $n = 10$.

Null Hypothesis: $H_0 \Rightarrow \mu = 0.700$ (The product is conforming to specifications)

Alternative Hypothesis; $H_1 \Rightarrow \mu \neq 0.700$

Test statistic: Under H_0 , the test statistic is: $t = \frac{\bar{x} - \mu}{\sqrt{s^2/n}} = \frac{\bar{x} - \mu}{\sqrt{s^2/(n-1)}}$

$$t = \frac{\sqrt{9} (0.742 - 0.700)}{0.040} = 3.15$$

$$\left[S^2 = \frac{s^2 \cdot n}{(n-1)} \right] \sim t_{(n-1)}$$

how to proceed further:- Here the test statistic 't' follows Student's - t distribution with $n-1 = 10-1 = 9$ degree of freedom. We will now compare this calculated value with the tabulated value of t for 9 d.f.

Case 1 If $t_{cal} = 3.15 > t_{tab.} \Rightarrow$ the value of t is significant.

$\Rightarrow \bar{x}$ differs significantly from μ

$\Rightarrow H_0$ may be rejected at this level of significance

$\Rightarrow$ The product is not meeting the specifications.

Case-2: $t_{cal} < t_{tab.}$

$\Rightarrow$ the value of t is not significant.

$\Rightarrow$ there is no significance difference between $\bar{x}$ and μ .

$\Rightarrow$ the deviation $(\bar{x} - \mu)$ is just due to fluctuations of sampling

$\Rightarrow$ The null Hypothesis H_0 may be retained at some (say 5%) level of significance

$\Rightarrow$ We may take the product confirming to specifications.

Q. The mean weekly sales of soap bars in departmental stores was 146.3 bars per store. After an advertising campaign the mean weekly sales in 22 stores for a typical week increased to 153.7 and showed a S.D. of 17.2. Was the advertising campaign successful?

Soln: - we are given $n = 22$, $\bar{x} = 153.7$,
 $s = 17.2$

Null Hypothesis: H_0 : The advertising campaign is not successful
: H_1 : $\mu = 146.3$

Alternative Hypothesis $\therefore H_1 : \mu > 146.3$ (Right tail)

Test Statistic : Under H_0 , the test statistic is :

$$t = \frac{\bar{x} - \mu}{\sqrt{s^2/(n-1)}} \sim t_{22-1} = t_{21}$$

$$t = \frac{153.7 - 146.3}{\sqrt{(17.2)^2/21}} = \frac{7.4 \times \sqrt{21}}{17.2} = 9.03$$

conclusion:

t_{tab} for 21 d.f. at 5% level of significance for single tailed test is 1.72. Since Calculated value is much greater than the tabulated value.

$$t_{tab} > t_{cal}$$

$\Rightarrow$ It is highly significant

$\Rightarrow$ We reject the null hypothesis.

Conclusion:

$\Rightarrow$ The advertising campaign was definitely successful in promoting sales.

Q. A random sample of 10 boys had the following I.Q.'s:
 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support
 the assumption of a population mean I.Q. of 100? Find a
 reasonable range in which most of the mean I.Q. values of
 samples of 10 boys lie.

Solⁿ. Null Hypothesis, H_0 : The data are consistent with the
 assumption of a mean I.Q. of 100 in the population i.e.
 $\mu = 100$.

Alternative Hypothesis: $H_1: \mu \neq 100$

Test Statistic, under H_0 , the test statistic is:

$$t = \frac{\bar{x} - \mu}{\sqrt{S^2/n}} \sim t_{n-1},$$

Where $\bar{x}$ and S^2 are to be computed from the sample values of I.Q.'s.

Calculation of Sample mean and S.D.

x	$x - \bar{x}$	$(x - \bar{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
<u>972</u>		<u>1833.60</u>

$$n = 10$$

$$\bar{x} = \frac{972}{10} = 97.2$$

$$S^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{1833.60}{9} = 203.73$$

$$|t| = \frac{|\bar{x} - \mu|}{\sqrt{s^2/n}} = \frac{|97.2 - 100|}{\sqrt{203.73/10}} = \frac{2.8}{\sqrt{20.37}} = \frac{2.8}{4.514}$$

$$= 0.62$$

Tabulated $t_{0.05}$ for $(10-1)$, i.e. 9 d.f. for two tailed test is 2.262.

Conclusions :- $t_{cal} < t_{tab.}$ for 95% confidence level and 9 d.f. H_0 may be accepted at 5% level of significance and we may conclude that the data are considered with the assumption of mean I.Q. values of sample of 10 boys will be given by :

$$\begin{aligned}\bar{x} \pm t_{0.05} S/\sqrt{n} &= 97.2 \pm 2.262 \times 4.514 \\ &= 97.2 \pm 10.21 \\ &= 107.41 \text{ and } 86.99\end{aligned}$$

Hence the required 95% confidence interval is
 $[86.99, 107.41]$.

Example:

A pesticide is packed into bags by a machine. A random sample of 15 bags is drawn and their contents are found to weight (in kg.) as follows:

50, 49, 52, 44, 45, 48, 46, 45, 49, 45, 47, 48, 41, 49, 46

Test if the average packing can be taken to be 51 kg.

Solution: Null Hypothesis, $H_0: \mu = 51$

Alternative hypothesis, $H_1: \mu \neq 51$

Under null hypothesis the test statistics is:

$$t = \frac{\bar{y} - \mu_0}{S / \sqrt{n}} = \frac{46.93 - 51}{2.76 / \sqrt{15}} = \frac{-4.07}{2.76 / 3.87} = \frac{-4.07}{0.71} = -5.69$$

It follows t-distribution with 14 degree of freedom.

The tabulated value of $t_{0.05}$ for 14 degree of freedom is 2.145. Here, $|t|$ is greater than tabulated value of t, so we reject the null hypothesis at 5% level of significance and we conclude that the average packing cannot be taken to be 51 kilograms.

Example:

Ten individual were chosen at random from a normal population and their heights were found to be in inches as 63, 63, 66, 67, 68, 69, 70, 70, 71 and 71. Test the hypothesis that the mean height of the population is 66 inches at 5 % level of significance.

Tabulated value= 2.26

Solution:

Null Hypothesis, $H_0: \mu = 66$

Alternative hypothesis, $H_1: \mu \neq 66$

Under null hypothesis the test statistics is:

$$t = \frac{\bar{y} - \mu_0}{S / \sqrt{n}} = \frac{67.8 - 66}{3.011 / \sqrt{10}} = 1.89$$

It follows t-distribution with 14 degree of freedom.

The tabulated value of $t_{0.05}$ for 9 degree of freedom is 2.26. Here, t calculated value is greater than tabulated value of t, so we accept the null hypothesis at 5% level of significance and we conclude that the mean height of the population is 66 inches.